Customer Integration Guide – SIP Registration over WebSocket (WSS)

This document describes the steps required to integrate your custom SIP client/softphone with the CloudConnect SIP platform using Secure WebSocket (WSS).

Step 1: Configure the SIP Account

Use the following account details in your application:

Parameter	Value
SIP Username	**102724701**
Authentication Username	**102724701**
SIP Password	**(Password provided separately)**
SIP Domain	**sip2.cloud-connect.in**
SIP Registrar	**sip2.cloud-connect.in:7065**
WebSocket URL	**wss://sip2.cloud-connect.in:7443/**
Transport	**WSS (Secure WebSocket)**
Registration Expiry	**3600 seconds**

> **Note:** The password is used only for SIP Digest Authentication and must not be sent in plain text.

Step 2: Establish the Secure WebSocket Connection

Before sending any SIP messages, establish a Secure WebSocket connection to:

```
wss://sip2.cloud-connect.in:7443/
```

A successful connection should return an HTTP **101 Switching Protocols** response.

Step 3: Send the Initial SIP REGISTER

Once the WebSocket connection is established, send the SIP REGISTER request.

Example:

```
REGISTER sip:sip2.cloud-connect.in:7065 SIP/2.0
Via: SIP/2.0/WSS <client-generated>.invalid;branch=<generated>;rport
Max-Forwards: 70
From: <sip:102724701@sip2.cloud-connect.in>;tag=<generated>
To: <sip:102724701@sip2.cloud-connect.in>
Contact: <sip:102724701@<client-generated>.invalid;transport=ws>
Call-ID: <generated>
CSeq: 1 REGISTER
User-Agent: <Your Application Name>
Expires: 3600
Content-Length: 0
```

Step 4: Validate the REGISTER Request

Please ensure the following:

- Use the complete SIP Username (`102724701`) in both the **From** and **To** headers.
- The REGISTER Request-URI should be:

  ```
  sip:sip2.cloud-connect.in:7065
  ```

- The `Via` , `branch` , `tag` , `Call-ID` , and `Contact` values should be generated by your SIP stack/application.
- Do **not** hardcode the server address in the `Via` or `Contact` headers.

Step 5: Handle SIP Authentication

After the initial REGISTER, your application should be capable of handling SIP Digest Authentication.

The expected SIP registration flow is:

```
REGISTER
      ↓
401 Unauthorized
      ↓
REGISTER (with Authorization header)
```

```
      ↓
200 OK
```

Your application must:

- Parse the `WWW-Authenticate` header.
- Calculate the Digest Authentication response using the provided SIP credentials.
- Send a second REGISTER request containing the `Authorization` header.

Step 6: Registration Success

Registration is considered successful only after receiving:

```
SIP/2.0 200 OK
```

Your application should then mark the extension as **Registered**.

Step 7: Troubleshooting

If registration is unsuccessful, please collect and share the following information:

- WebSocket connection status
- Complete outbound REGISTER request
- Complete inbound SIP response
- SIP Response Code (e.g., 401, 403, 404)
- Timestamp
- Call-ID
- CSeq

This information will help us analyze the registration flow more efficiently.

Important Notes

- Use **WSS** only. Do not attempt registration using plain `ws://`.
- WebSocket transport should connect to:

  ```
  wss://sip2.cloud-connect.in:7443/
  ```

- The SIP Registrar should be:

  ```
  sip2.cloud-connect.in:7065
  ```

- Ensure the SIP Username, Authentication Username, and Password exactly match the credentials provided by CloudConnect.
- The password should never be transmitted in plain text. It is used only to generate the SIP Digest Authentication response.

If you continue to receive a SIP error (such as **403 Forbidden**) after implementing the above steps, please share the SIP logs (REGISTER request and server response) with us for further analysis.